

Ratios with Tape Diagrams and Double Number Lines

Answer Key

Grade 6

Quick Answers

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|---|--|
| 1. 8 L | 6. (a) 1 pound = 1.5, (b) 10 pounds = 15 |
| 2. (a) 4 minutes = 20, (b) 6 minutes = 30 | 7. 63 |
| 3. 20 | 8. 35 cm |
| 4. 10 cups of flour = 4, 15 cups of flour = 6, 20 cups of flour = 8 | 9. (a) 10 minutes = 3, (b) 50 minutes = 15 |
| 5. 20 | 10. cups of yogurt = 8, — |
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What is verified

9 of 10 problems fully machine-checked against SymPy.

— marks an answer only you can judge — problem 10. The worked solution states what a correct response must contain.

Curriculum

Level: Grade 6

6.RP.A.2 – *problems 1, 6*

6.RP.A.3 – *problems 2, 3, 4, 5, 7, 8, 9, 10*

Difficulty 1–4 of 5 across 16 tagged checks.

Common wrong answers

3. *If they got 17.5: divided the 35 dollars by the 2 people instead of the 7 ratio parts*
5. *If they got 60: treated the 12 extra stickers as the value of one box, answering 5 times 12*
8. *If they got 63: found the length of the long piece (63 cm) instead of the difference*
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1. Tape diagram: 2 juice boxes and 3 water boxes, each box worth 4 liters. How many liters of juice?

Solution: The juice tape has 2 boxes and every box stands for 4 liters, so juice = $2 \times 4 = 8$ liters. 8 L

2. Hose fills 10 liters every 2 minutes. Complete the double number line at 4 and 6 minutes.

Solution: Each 2-minute step adds 10 liters, so the marks go up by 10:

(a) at 4 minutes: $10 \times 2 = 20$ liters.

20

(b) at 6 minutes: $10 \times 3 = 30$ liters.

30

3. Maria and Josh share 35 dollars in the ratio 4:3. How much does Maria receive?

Solution: The tape has $4 + 3 = 7$ equal boxes for 35 dollars, so one box is worth $35 \div 7 = 5$ dollars. Maria's tape has 4 boxes: $4 \times 5 = 20$ dollars.

20

4. Sugar to flour is 2 to 5. Complete the sugar row for 10, 15, and 20 cups of flour.

Solution: Each flour step of 5 pairs with a sugar step of 2. For 10 cups of flour the multiplier is 2, so sugar = $2 \times 2 = 4$.

4

For 15 cups the multiplier is 3: sugar = $2 \times 3 = 6$.

6

For 20 cups the multiplier is 4: sugar = $2 \times 4 = 8$.

8

5. Ana and Ben have stickers in the ratio 5:2, and Ana has 12 more. How many does Ana have?

Solution: Ana's tape shows $5 - 2 = 3$ more boxes than Ben's, and those 3 extra boxes hold the 12 extra stickers, so one box = $12 \div 3 = 4$ stickers. Ana has 5 boxes: $5 \times 4 = 20$ stickers.

20

6. Apples cost 6 dollars for 4 pounds. Find the money marks for 1 pound and 10 pounds.

Solution: (a) Divide both quantities by 4 to reach one pound: $6 \div 4 = 1.5$ dollars per pound.

1.50

(b) Multiply the one-pound mark by 10: $1.5 \times 10 = 15$ dollars.

15

7. Fiction to nonfiction is 7:5, and the 5 nonfiction boxes hold 45 books. How many fiction books?

Solution: One box is worth $45 \div 5 = 9$ books. The fiction tape has 7 boxes: $7 \times 9 = 63$ books.

63

8. A 91-cm rope is cut in the ratio 9:4. How much longer is the long piece?

Solution: The full tape has $9 + 4 = 13$ boxes for 91 cm, so one box is $91 \div 13 = 7$ cm. The long piece has $9 - 4 = 5$ more boxes than the short piece, so the difference is $5 \times 7 = 35$ cm.

35 cm

(A common slip is reporting the long piece itself, $9 \times 7 = 63$ cm, instead of the difference.)

9. 27 km in 90 minutes at a steady pace. Find the distance for 10 minutes and for 50 minutes.

Solution: (a) Divide both quantities by 9 to get a 10-minute step: $27 \div 9 = 3$ km every 10 minutes. 3

(b) 50 minutes is five 10-minute steps: $3 \times 5 = 15$ km. 15

10. Berries to yogurt is 3:2; the batch uses 12 cups of berries. How many cups of yogurt? Explain how the tape helped.

Solution: The 3 berry boxes hold 12 cups, so one box is worth $12 \div 3 = 4$ cups. The yogurt tape has 2 boxes: $2 \times 4 = 8$ cups. 8

Explanation (instructor judges): Full credit for one sentence that finds the value of one box from the berries ($12 \div 3 = 4$ cups per box) and then counts the 2 yogurt boxes ($2 \times 4 = 8$ cups) — the diagram turns the ratio into equal boxes you can count.